DO NOW — RECALL

Spiral model and Rapid Application Development

Name: _____ Class: _____ Date: _____

1. Name a key XP programming practice.

2. How does an Agile sprint help with changed requirements?

3. What is a prototype?

4. Why might risk analysis be valuable before developing an uncertain hardware-dependent system?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

For example pair programming, TDD or continuous integration.

2. Model answer

It lets the team adjust the next iteration using feedback.

3. Model answer

An early model used to explore and test an idea.

4. Model answer

It can reveal technical feasibility issues before expensive implementation.

Vocabulary — know these terms

Spiral model — Iterative methodology driven by risk identification and evaluation.

risk analysis — Evaluate probability/impact of potential problems.

RAD — Rapid Application Development using frequent prototypes and feedback.

prototype — Early working or mock representation for evaluation.

stakeholder — Person or group affected by or using the system.

NEW LEARNING

Read, annotate and explain

Spiral focuses on risk

Each cycle plans objectives, identifies or resolves major risks, develops/tests and reviews with stakeholders.

RAD is prototype-led

Users evaluate quickly produced models. Frequent prototyping accelerates understanding and can reveal unclear requirements.

Compare suitability

Spiral can suit large/high-risk systems but increases management cost; RAD relies on accessible users and skilled teams and may compromise architecture if rushed.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A satellite control project has major safety and technology uncertainties. Recommend a method.

Predict first

Spiral is appropriate because major technical and safety risks need assessment at each cycle.

Model and explain

It demands experienced risk assessment and carries greater time/management overhead than simpler methods.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State one key characteristic and one disadvantage of both Spiral and RAD.

Question 2

A transport company is building an experimental driverless control system. Compare Spiral with RAD and justify a choice.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A booking website must launch rapidly and users are available daily. Explain three benefits and two risks of RAD.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Spiral development and RAD are identical because both can repeat stages.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.